Trova: morning start guide

Grok Hackathon Pavia · Wed 4 Nov 2026 · Track A: CTRL+F: SEA (SEA Vision)

What we are building, in 30 seconds

You ask for an object, typed, spoken (English or Italian) or as a phone photo. Trova finds it on the table and the cobot points at it. Then it compares the table with an approved state and reports what was added, removed or moved. That is **line clearance**, the check SEA Vision sells to pharma packaging lines. **Nothing is trained:** Grok already knows what objects are.

The code is already written. It runs today in demo mode on a simulated table and arm (25 tests pass). Your job is to connect the real camera and arm, switch on live Grok, tune on the real objects, measure, and rehearse.

The first three things at 09:00

1. **Claim Track A** with the organisers. Only 2-3 teams can take it.
2. **Ask for the arm:** UFactory xArm5 first, UR5e second. Get its IP address and the slot calendar.
3. **Get the Grok API key** into the file `.env`. The badge in the app turns from MOCK to LIVE.

Who does what (write the names in)

Role	Owns	Name
P1 Arm	Connecting and calibrating the cobot; STOP; drives the arm in the demo	
P2 Camera and vision	Mounting the camera, empty-table capture, object boxes, markers	
P3 Grok	API key, prompts, speed of each find	
P4 What changed and tests	Test set, numbers against baselines, README results	
P5 Demo, voice, pitch	Questions for SEA Vision, Grok Voice test, slides, backup video	

Golden rules. Staff present for every arm move. STOP always within reach (button, space bar, or say "stop" / "fermo"). Grok never moves the arm by itself: every move goes through code with a safety box. Never commit `.env`. Feature freeze 02:00, code freeze Thursday 12:00, demo Thursday 16:00.

Before you leave home: bring

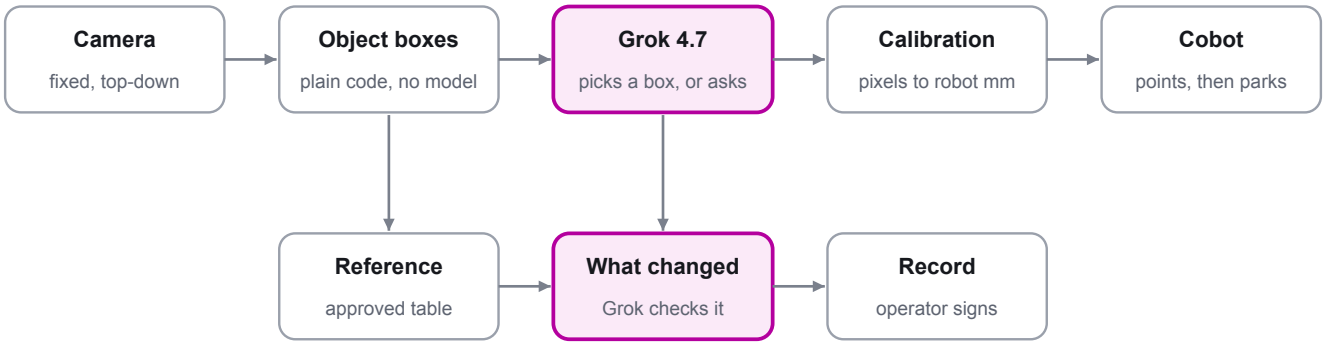
- Laptop with its charger, and the repo already installed (see section 3).
- **USB-C to Ethernet adapter and an Ethernet cable.** The arms connect by cable and MacBooks have no Ethernet port. Without this we cannot talk to the arm.
- A phone, a clamp or small tripod, and a USB cable: a second, sharper camera (iPhone works as a webcam).
- 4 printed ArUco markers, 5 cm wide, and tape. Print them from the app: Setup, Print markers.
- A phone hotspot (venue Wi-Fi carries 100 people) and an extension lead.

1. The hardware

Ask the organisers for

Hardware	Choice	Why
UFactory xArm5	**1st**	Official Python SDK, no pendant setting to change. 700 mm reach covers the table.
Universal Robots UR5e	**2nd**	Best reach (850 mm). Staff must enable Remote Control on its pendant, which may need an admin password.
Elite Robots CS66	3rd	914 mm reach, but our least-tested driver.
UR3e, Elite CS63	Avoid	Reach too short for a full table (500 / 624 mm).
Camera 2 (the station's)	Yes	The top-down camera of the Track A station.
Unitree Go2 / Go1 dog	Skip for now	Optional scenario. Photos of the rooms are enough for the simulated dog; ask for a Go2 slot only if the core demo already works.

2. How the setup works



Top row: a find. Bottom row: a line-clearance check. Typing, a phone photo or Grok Voice all start the same flows.

- **Camera.** Fixed above the table, looking down. Once per arm slot we photograph the empty table. Anything that differs from that photo is an object and gets a numbered box. Shadows and light changes are ignored.
- **Grok.** It receives the photo with numbered boxes, close-ups of each object and the request. It answers with a box number, "not found", or a question when two objects fit ("the blue pen or the black marker?").
- **Calibration.** The arm tip is moved by hand (freedrive) onto 4-6 spots on the table, and each time you click the tip in the live view. Trova then knows which camera pixel is which robot position, in millimetres.
- **The arm.** It receives one target: x, y, about 15 cm above the table, pointing down. The robot works out its own joint angles. It moves in a straight line, slowly, inside a safety box, and parks out of the camera's view before each new photo. We never program left, right or forward: calibration takes care of directions.
- **What changed.** Save a reference of the approved table. Later, Trova compares object by object (position, colour, shape), Grok confirms each change and says why it matters, and the operator signs a record.
- **Voice.** On the laptop, press Talk to Trova and speak English or Italian. It asks before the arm moves; the word "stop" halts the arm immediately, without waiting for Grok.

The objects must: look different from the table (pale-on-pale and shiny metal can be missed), not touch each other (touching objects merge into one box), and sit inside both the camera view and the arm's reach. The station's 20-30 objects come from SEA Vision; we do not need to bring any.

3. How to make it: the morning, step by step

Before 09:00 (on the train): install and try the demo

```
git clone https://github.com/Aman-AI-Guy/Grok_Hackathon && cd Grok_Hackathon
python3 -m venv .venv && .venv/bin/pip install -e ".[dev,robots]"
cp .env.example .env
.venv/bin/python -m trova          # then open http://localhost:8000
```

Time	Who	What
09:00	P1 + P5	Claim Track A. Ask for the xArm5 (or UR5e), its IP, and the arm slot calendar.
09:00	P3	Put the API key in `.env`, restart the app: badge shows LIVE.
09:00	P5	Ask Claudio Peroni the questions in section 4.
09:15	P2	Mount the camera top-down; set `TROVA_CAMERA`; press Capture empty table; tune until every object has exactly one box; record 10 minutes of station video for offline work.
09:15	P3	Run `experiments/grok_bbox_test.py` on station photos; compare the speed of grok-4.7 and the faster model.
09:30	P4	Start recording test cases: ask for an object, then press correct or wrong.
First arm slot	P1 (+ P2)	Connect, first slow move, test STOP, calibrate (steps below).
13:00 target	Everyone	**First real find:** real camera, live Grok, real arm pointing.
Afternoon	Everyone	Measure, tune, live voice test, rehearse. Record the backup video tonight.

The first arm slot (P1, with museum staff)

1. Plug in the Ethernet adapter. In the Mac's network settings give it a manual IP on the arm's subnet, for example 192.168.1.50, mask 255.255.255.0, and **no router**, so the internet stays on Wi-Fi. Ping the arm.
2. UR or Elite only: staff enable Remote Control on the pendant. On every arm: set the TCP offset of the printed tool, otherwise positions refer to the flange, not the tip.
3. Fill in `.env` (below), restart the app, make the first move at low speed, and press STOP in the middle of a move to check that it halts and holds.
4. Freedrive on, put the tip on the table at 4-6 spread-out spots and click the tip each time (Calibrate with arm). Then touch each ArUco marker and record it, so later recalibration takes seconds.

```
TROVA_ARM=xarm:192.168.1.xxx        # or ur:<ip>  or  elite:<ip>
TROVA_ARM_SPEED_MM_S=50
TROVA_ARM_Z_MM=150                  # tip height: table surface + 3-5 cm, measure it
TROVA_ARM_LIMITS=xmin,xmax,ymin,ymax,zmin,zmax   # moves outside are refused
TROVA_ARM_HOME_MM=x,y,z             # parked pose, out of the camera's view
```

A real arm refuses to move until it has been calibrated with the arm itself. If the arm misbehaves, keep working with `TROVA_ARM=sim` and come back in the next slot.

4. Who to talk to

Who	Ask them about	When
Organisers: Luca Bianchi, Federico Pellegrini (hosts)	Claim Track A. Prizes per track or overall? The common judging rubric. Pitch length. API credits.	09:00 kick-off
Museum hardware staff	Which arm is arm 2 and its IP. Remote Control and TCP offset. Arm slot calendar, overnight too. May we mount our camera and tape markers on the table? Dog access, later.	09:00, then every slot: staff must be present for every arm move
Claudio Peroni, SEA Vision (mentor and juror)	The object list. Five requests like the jury's. How close the tip must be to count as found. How find time is measured. Is the lighting fixed? How far must an object move to count as moved?	Wednesday and Thursday
Michele Antolini, SEA Vision (mentor)	A technical look at the demo before the pitch.	Thursday
Ralph Johnson, SpaceXAI	Grok API and Grok Voice: speed settings, voice latency, structured outputs.	Thursday: 09:00 talk, then at the tables
Inside the team	Arm: P1 · Camera: P2 · Grok: P3 · Tests and numbers: P4 · Voice and pitch: P5	Always

One person (P5) collects everyone's questions for Claudio Peroni, so he is asked once, not five times. He is also on the jury: show him progress, not problems.

5. How to understand everything in 20 minutes

1. **Watch the 101-second concept video**: `media/trova_concept.mp4` in the repo. It shows every moment of the demo: find by words, by photo, by voice, what changed, the record and the dog.
2. **Play with the demo** on your laptop (demo mode is the default): type "the red one" and press Find; press + add, Save reference, Check now; open the record.
3. **Read the README**, sections "What Grok does" and "Layout": they map every file to what it does.
4. **The full team plan** (roles, schedule, risks, market): claude.ai/artifact/Xc5nfd5Q41HmeHHwScSxmb

Words you will hear

Term	Meaning
Mock mode	No API key: Grok answers are simulated (colour words only). The badge says MOCK.
Empty table	Photo of the table without objects. Everything different from it is an object.
Object boxes	Numbered rectangles that plain code draws around each object. Grok picks a number.
Reference	The approved state of the table that "what changed" compares against.
Calibration	4-6 clicks that map camera pixels to robot millimetres.
TCP	The tool tip of the arm. Its offset must be set, or positions refer to the flange.
Remote Control	Pendant setting on UR and Elite that lets a laptop send moves.
Freedrive	The arm goes limp so a person can move it by hand. Calibration only, staff present.
Jury mode	Correct or wrong after each find. It saves test cases and keeps the score.